

■ Future Scope, Limitations & Conclusion

Project summary, known limitations, improvements, and references

Project Conclusion

The Scuba Diving Underwater Health Monitoring System using Li-Fi Technology successfully demonstrates a reliable, real-time solution for monitoring diver health in underwater environments. The system integrates ESP32 microcontrollers, biomedical sensors, and Li-Fi optical communication to continuously track heart rate, SpO2, and body temperature, transmitting this data to the surface team in real time.

The experimental results confirmed the effectiveness of Li-Fi communication over conventional RF methods underwater. The emergency mechanisms — air balloon inflation, surface buzzer, and panic switch — were validated successfully. The system achieves its core objectives of diver safety, reliable communication, and timely emergency response.

Key Achievements

- Real-time health monitoring: Heart rate ($\pm 2\text{--}3$ BPM), SpO2 ($\pm 2\%$), Temperature ($\pm 0.2^{\circ}\text{C}$)
- Li-Fi communication: $<100\text{ms}$ latency, near-zero packet loss in clear water
- Emergency response: Buzzer 0.5s | Relay 1.2s | Air motor balloon 1.8s
- Panic switch: Manual SOS in <0.3 seconds
- Cloud dashboard: Live ThingSpeak graphs for all three parameters
- Full system validation through controlled underwater testing

Known Limitations

Limitation	Details
Limited Li-Fi Range	Effective only up to 2–3 meters; requires line-of-sight between LED and photodiode.
Water Turbidity Sensitivity	Muddy, polluted, or particle-rich water reduces light penetration and signal quality.
Ambient Light Interference	Strong sunlight or external light sources near the receiver can corrupt data.
Battery Life	High-intensity LED operation and continuous sensor sampling reduce battery duration.
Single Diver Only	Current design supports one diver at a time; no multi-diver simultaneous monitoring.
Sensor Placement	MAX30102 accuracy depends on firm contact; loose placement causes reading errors.

Future Scope & Improvements

1. Extended Li-Fi Range

- Use high-power LEDs or laser-based Li-Fi transmitter with optical focusing lens
- Implement hybrid communication: Li-Fi for short range + acoustic signalling for backup
- Target: extend range from 3m to 10–20m underwater

2. Multi-Diver Support

- Assign unique Li-Fi frequencies or time-division multiplexing for each diver
- Surface dashboard shows all divers' vitals simultaneously
- Centralized monitoring for dive teams of 5–10 divers

3. AI-Based Predictive Alerts

- Train ML model on diver health patterns to predict emergencies before thresholds are breached
- Pattern recognition for fatigue, decompression sickness early signs
- Integrate TensorFlow Lite on ESP32 for on-device inference

4. Additional Sensors

- Depth sensor (pressure sensor) for dive depth monitoring
- Breathing rate sensor (respiratory monitoring)
- Orientation/IMU sensor (detect diver unconsciousness by posture)
- CO2 sensor for air quality inside breathing apparatus

5. Waterproofing & Packaging

- IP68 certified waterproof enclosures for all electronics
- Neoprene bands for secure sensor attachment to diver's wrist/finger
- Sealed battery pack with underwater charging capability

6. Mobile App Integration

- iOS/Android app for real-time monitoring by surface team
- Push notifications when any parameter crosses threshold
- GPS location logging + dive history recording

Applications

- Professional Scuba Diving: Commercial deep-sea construction and maintenance divers
- Underwater Rescue Missions: Monitor rescue diver health during search and recovery
- Marine Research & Oceanography: Safety for scientists during long underwater experiments
- Defence & Naval Diving: Secure, interference-free monitoring for military divers
- Sports Diving: Recreational diver safety in open water and cave diving

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